THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The impulse of a force of 20 N acting for 0.5 s is: 1

(a) 0.5 N s **(b)** 20 N s **(c)** 40 N s **(d)** 10 N s

2. A car of mass 1200 kg is moving at 20 m/s. The force needed to stop it in 6 s is: 1

(a) 240 N **(b)** 7200 N **(c)** 4000 N **(d)** 2000 N

3. An astronaut in orbit feels weightless because: 1

(a) he is far from Earth **(b)** gravity is zero there

(c) of no air **(d)** he and the spacecraft are in free fall

4. A cricket ball of mass 0.15 kg moving at 20 m/s is hit back at 20 m/s. The change in momentum is: 1

(a) 3 kg m/s **(b)** 1.5 kg m/s **(c)** 6 kg m/s **(d)** 0

5. A force of 15 N acts on a 3 kg body at rest. The distance covered in 2 s is: 1

(a) 5 m **(b)** 30 m **(c)** 20 m **(d)** 10 m

6. The weight of a 4 kg body at a place where g = 9.8 m/s squared is about: 1

(a) 2.45 N **(b)** 9.8 N **(c)** 4 N **(d)** 39.2 N

7. The recoil of a gun is explained by: 1

(a) Newton's third law **(b)** first law

(c) gravitation **(d)** friction

8. A 5 kg body at rest is acted on by a force of 20 N. Its velocity after 2 s is: 1

(a) 10 m/s **(b)** 8 m/s **(c)** 4 m/s **(d)** 40 m/s

9. The weight of a body is 98 N on Earth (g = 9.8). Its mass is: 1

(a) 98 kg **(b)** 1 kg **(c)** 10 kg **(d)** 9.8 kg

10. The velocity of recoil of a gun is small because the gun has a: 1

(a) short barrel **(b)** large mass **(c)** small mass **(d)** small force

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the force of friction not depend much on the area of contact? 2

12. State why a cricketer lowers his hands while catching a fast ball. 2

13. A truck and a car move with the same speed. Which needs a greater force to stop in the same time? Why? 2

14. A 2 kg body is pulled by 10 N against a friction of 4 N. Find the acceleration. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the origin of the three Newton's laws using the example of a football kick. 3

16. A 2 kg block is on a table of friction 5 N. Find its acceleration when pulled by 15 N and its velocity after 3 s. 3

17. Explain why the astronaut floats in an orbiting spacecraft though gravity acts on him. 3

OR

A body moves with uniform velocity 5 m/s. A force of 10 N then acts for 2 s in its direction on a body of mass 4 kg. Find the final velocity and the change in momentum. 3

18. A body of mass 2 kg is thrown upward with 20 m/s. Find the force of gravity on it, its acceleration at the top and the momentum at the start. 3

OR

A body is acted on by forces F1 = 5 N and F2 = 5 N at 180 degrees to each other. What is the state of the body? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss why the forces of action and reaction in the case of a book resting on a table do not cancel, and identify the pairs. 5

OR

Compare the motion of a heavy and a light ball dropped from the same height in the presence and absence of air, with the help of the second law. 5

20. Explain the concept of impulse with the example of a ball hitting a wall. A 0.4 kg ball hits a wall at 15 m/s and rebounds at 10 m/s in 0.02 s. Find the impulse and the force. 5

OR

Explain why wearing seat belts, crash helmets and using sand beds in long jump pits reduce injury, using F = change in momentum/time. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND